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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 4: Handoff Simulation with Hysteresis

Program Code:

```
import numpy as np
import matplotlib.pyplot as plt
# Signal strength from two base stations
t = np.arange(0, 20, 1)
bs1 = np.linspace(-50, -100, 20)  # Base Station 1
bs2 = np.linspace(-100, -50, 20)  # Base Station 2
hysteresis = 5
serving_bs = "BS1"
handoff_points = []
# Handoff simulation
for i in range(len(t)):
    if serving_bs == "BS1" and bs2[i] > bs1[i] + hysteresis:
        serving_bs = "BS2"
        handoff_points.append(i)
    elif serving_bs == "BS2" and bs1[i] > bs2[i] + hysteresis:
        serving_bs = "BS1"
        handoff_points.append(i)
    print("Time:", t[i], "Serving Base Station:", serving_bs)
print("\nHysteresis:", hysteresis, "dB")
print("Total Handoffs:", len(handoff_points))
```

```
print("Handoff Time:", t[handoff_points] if handoff_points else "No handoff")

print("\nResult: Handoff simulation completed successfully.")

# Plot graph

plt.plot(t, bs1, "b-o", label="Base Station 1")

plt.plot(t, bs2, "g-o", label="Base Station 2")

for point in handoff_points:

    plt.axvline(t[point], color="red", linestyle="--",

            label="Handoff" if point == handoff_points[0] else "")

plt.xlabel("Time")

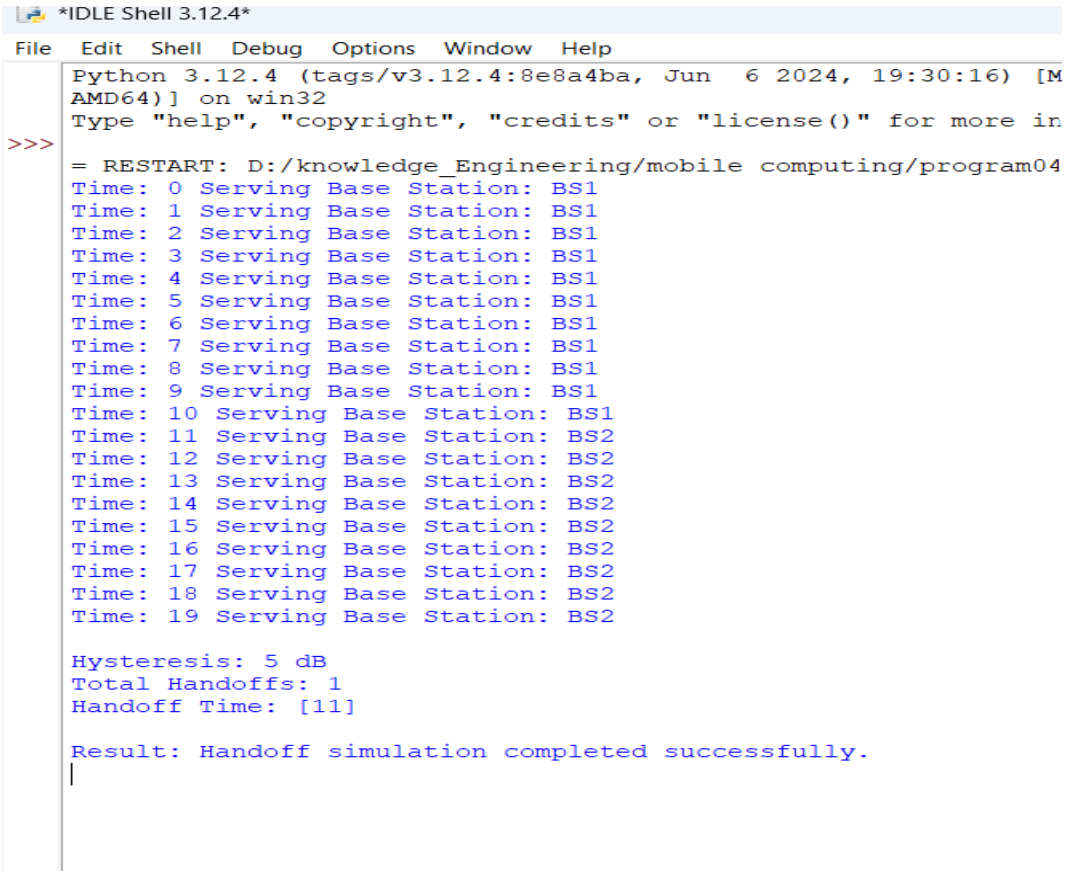
plt.ylabel("Signal Strength (dBm)")

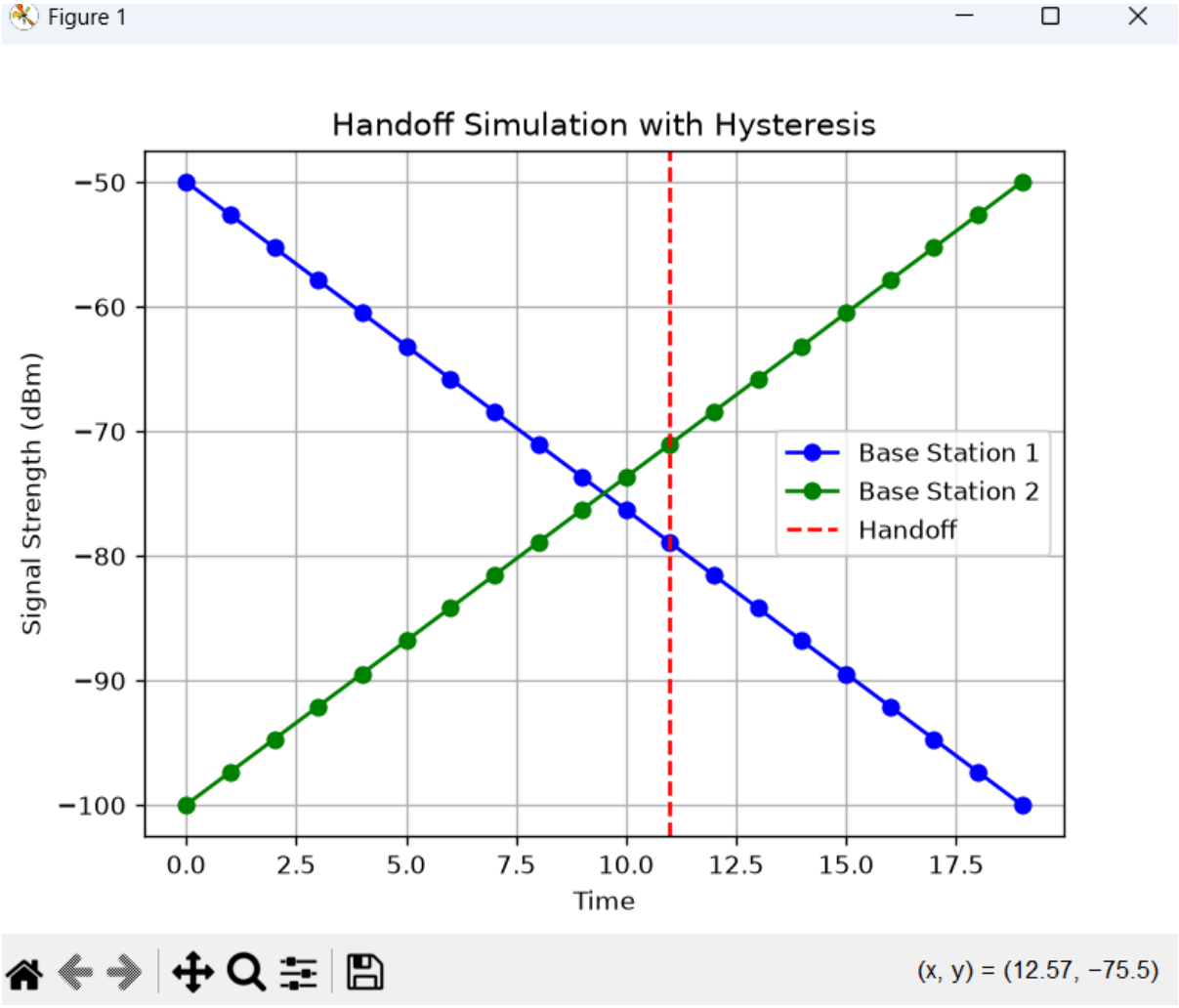
plt.title("Handoff Simulation with Hysteresis")

plt.legend()

plt.grid(True)

plt.show()
```




Figure 1
Handoff Simulation with Hysteresis
Signal Strength (dBm)
−50
−60
−70
−80
−90
−100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time
Base Station 1
Base Station 2
Handoff
(x, y) = (12.57, −75.5)